Winter Semester 2025/2026
Instructor: Barbara Pertold–Gebicka
Teaching Assistant: Kai Wang

Home Assignment 4

Deadline: December 16, 2025, 23:59

Dear students,

You are encouraged to work in groups of two people. In that case, hand only one solution with the names of both participants on the first page. In case you cannot find any colleague to form a group with, you can write an email to *kai.wang@fsv.cuni.cz* and I will match you with somebody. Submit your solution electronically via SIS. Pack all electronically submitted files into a zipped folder named **"Surname1_Surname2_HA1"**.

The solution of the first problem should be handwritten and might be submitted on paper to the mailbox No.14 (Barbara Pertold-Gebicka) or scanned and submitted electronically. The solution of the second problem can be written in any text-editing software of your choice (e.g., MS Word, LATEX) and should be submitted electronically. Attach both R "log" file with results and "rmd" file with the programming code to your assignment. If using another statistical software, attach the corresponding file(s). Use comments in your programming code to describe in few words what each command means. In case of problems with R, consult the official online manual and help module.

Keep in mind that not only the correctness of the results is assessed, but also the text editing quality is taken into consideration. Answer only what you are asked for.

Note that the use of **Artificial Intelligence (AI)** tools is only allowed to improve your writing style and grammar or to generate inspiration. However, treat the AI tools' output carefully and with caution, respecting the principles of academic integrity and the requirement to produce your own original work that is based on reliable and verifiable sources of information. Simple copying of the text generated by the AI tools is not permitted. Consult the full text of AI guidelines here.

Problem 1

Did you know that one can obtain OLS estimator using the maximum likelihood estimation technique? Your task is to do so. Remember to submit hand-written solution!

1. Start by writing down a simple linear regression model: $y_i = \beta_0 + \beta_1 x_i + u_i$.
2. What is the probability that an observation i has the value of the dependent variable equal to y_i given the observed value of the dependent variable, x_i? In other words, what is the density of y given x?
3. Use your derivations in (b) to build a likelihood function for observation i.
4. Log-linearize the likelihood function and sum it over all observations.
5. Maximize the log-likelihood function with respect to β_1. Did you arrive at any familiar formula? Comment.

Problem 2

Financial institutions face significant challenges in managing consumer credit risk. To support better decision-making in credit risk management, you are asked to analyze the *Ecox2_HW4_data* dataset. The dataset contains 30,000 credit card customers and includes demographic characteristics, credit limits, repayment history, bill amounts, payment behavior, and an indicator of whether the client defaulted in the next period. Your goal is to identify the factors associated with default risk, to estimate an appropriate econometric model, and to evaluate its predictive performance.

The solution to this problem should be submitted in the form of a mini-project report (max 5 pages, preferably less, including graphs and tables) with an introduction, main text, and conclusion. **Do not provide the solution in numbered or bullet points**, but write a consistent text with equations, graphs, and tables. Do not include R-code in the text; submit it as an attachment.

Treat the points below as a suggestive checklist.

1. Identify the variable indicating whether a client defaulted in the next month. Describe the nature of this variable (binary or other). Discuss which variables appear to be potentially useful predictors of default and which may be irrelevant for the analysis.
2. Examine the dataset for missing values, inconsistent categories, or outliers. Explain how you handle any detected issues and justify your pre-processing decisions.
3. Produce summary statistics and correlations for numeric variables (e.g., using `corrplot`). For categorical variables, create bar charts (e.g., using `ggplot`) to illustrate default rates across categories such as education, marriage status, gender, or repayment status.

4. Based on the nature of the dependent variable, choose appropriate estimation method(s). Explain the advantages and disadvantages of the chosen method(s). Clearly write out your model and discuss the expected signs of the key parameters.

5. Estimate the model. Evaluate the direction and significance of the estimated coefficients. Discuss whether the signs of key parameters are consistent with economic reasoning. Experiment with alternative sets of predictors and compare goodness-of-fit across different specifications and different econometric methods (e.g., using pseudo-R^2, percent correctly predicted, AIC, or BIC).

6. Report the average partial effects (APE) or partial effects at the average (PEA) for the model that you trust the most. Discuss the magnitude of the marginal effects and interpret their economic meaning. Highlight any surprising results.

7. Split the dataset into training and test sets. Use the training data to estimate model parameters and the test data to predict credit default. Evaluate the reliability of predictions using a confusion matrix and evaluate accuracy, sensitivity, and specificity. If you use a Logit model, you may produce an ROC curve and compute the area under the curve.

8. Based on your empirical results, propose recommendations for improving credit risk management. Discuss how credit limits, approval policies, or monitoring strategies may be adjusted according to the characteristics most strongly associated with default risk.

Variables definition

Response variable

- **default.payment.next.month**: Binary default indicator (1 = default, 0 = no default).

Key explanatory variables

- **X1**: Credit limit (NT dollars).
- **X2**: Gender (1 = male; 2 = female).
- **X3**: Education (1 = graduate school; 2 = university; 3 = high school; 4 = other).
- **X4**: Marital status (1 = married; 2 = single; 3 = others).
- **X5**: Age (years), no bigger than 130.

Repayment status variables (X6–X11) These variables record payment delays from April to September 2005. Coding scheme:

- -1 = paid duly,
- 0 = minimum payment,
- 1 = delay for one month,
- 2 = delay for two months,
- ...
- 8 = delay for eight months,
- 9 = delay for nine months or more.

Bill amounts (X12–X17) Monthly bill statements for the six months from April to September 2005 (NT dollars).

Repayment amounts (X18–X23) Amounts paid in each of the same six months (NT dollars).

Source Dataset available at: `https://archive.ics.uci.edu/dataset/350/default+of+credit+card+clients`.